Regime-Adaptive Cross-Sectional Momentum in Cryptocurrency Markets: A Walk-Forward Validated Long/Short Strategy with Derivatives-Based Quality Gates

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Abstract

We propose RACM, a three-layer daily-frequency long/short momentum strategy for six major cryptocurrency assets (BTC, ETH, SOL, XRP, DOGE, LINK). The first layer ranks assets by volatility-adjusted cross-sectional momentum (60-day and 90-day lookbacks) to construct long-short portfolios rebalanced every eight hours. The second layer detects market regimes using a composite danger score derived from a 110-day moving average crossover, 30-day return skewness, and 45-day drawdown, scaling positions between $-0.7\times$ and $1.5\times$ across four discrete states. The third layer applies cryptocurrency-specific quality gates based on funding rate, open interest change, and liquidation cascade z-scores to suppress exposure during structurally dangerous conditions. Position sizing follows a fractional Kelly criterion ($f = 0.15$) with a $3.0\times$ cap and equity-based drawdown control.

We evaluate the strategy using a strict walk-forward protocol: 3-month training / 1-month testing, rolling monthly for 45 folds (2021–2024 in-sample). The strategy achieves an 87% fold win rate (39/45), a walk-forward score (WFS) of 367% annualized, and +121% during the 2022 bear market (BTC: −65%). On a genuinely out-of-sample 2025 period with no parameter adjustment, the strategy returns +93% (BTC: −7%, MaxDD: −13.6%). Permutation tests (1,000 iterations, three variants) yield $p = 0.000$. However, a t-test cannot reject WFS $> 300\%$ ($p = 0.206$), and IS-to-OOS degradation is 78%. We discuss these limitations transparently, arguing that the statistically defensible claims are the win rate exceeding 50% ($p < 0.0001$) and positive OOS alpha in a down market.

Keywords: cryptocurrency, cross-sectional momentum, regime detection, walk-forward validation, long-short strategy, Kelly criterion

1 Introduction

Cryptocurrency markets present unique challenges for quantitative strategies. Annual volatility of 80–120% (approximately 4–5× that of equities) makes simple buy-and-hold

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approaches extremely risky: Bitcoin declined 65% in 2022, while the broader altcoin market experienced drawdowns exceeding 90%. This motivates the search for strategies that perform in both bull and bear environments.

Short-term price prediction in cryptocurrency markets has proven extremely difficult. Machine learning classifiers, order book imbalance features, and derivatives-based directional signals all fail to generate statistically significant alpha after transaction costs at intraday to daily frequencies.[1] This negative result motivates a fundamental shift: from *prediction* to *risk management*.

Our contribution is threefold. First, we apply the cross-sectional momentum framework of Moskowitz et al. [2012] to a cryptocurrency universe, demonstrating that vol-adjusted momentum rankings generate a modest but persistent long-short spread. Second, we introduce a regime detection system that scales position sizes based on a composite danger score, enabling safe use of leverage that would otherwise be catastrophic in bear markets. Third, we incorporate cryptocurrency-specific quality gates derived from perpetual futures market microstructure (funding rates, open interest dynamics, and liquidation cascades) as a risk management overlay.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the data and methodology. Sections 4 and 5 present in-sample and out-of-sample results, respectively. Section 6 provides robustness tests. Section 7 discusses limitations. Section 8 concludes.

2 Related Work

2.1 Momentum in Traditional Assets

Jegadeesh and Titman [1993] established that buying past winners and selling past losers generates abnormal returns in equities. Moskowitz et al. [2012] extended this to time-series momentum across asset classes, showing that past 12-month returns predict future 1-month returns in futures markets. We adapt this framework to a cross-sectional cryptocurrency context with substantially shorter lookback periods.

2.2 Momentum in Cryptocurrencies

Grobys et al. [2020] document momentum profits in cryptocurrency markets, though with concerns about survivorship bias and market microstructure. Liu et al. [2022] construct cryptocurrency factor models analogous to Fama-French. Our work differs by employing a long-short construction on a fixed 6-asset universe with explicit regime conditioning.

2.3 Regime Detection

Ang and Timmermann [2012] provide a comprehensive framework for regime-switching models in financial markets. Our approach uses a simpler rule-based composite score rather than a hidden Markov model, prioritizing interpretability and avoiding the estimation complexity of latent state models in highly volatile cryptocurrency markets.

[1] We tested LSTM, Transformer, Random Forest classifiers on 15-minute to 3-hour prediction horizons, as well as order flow imbalance (OBI) and derivative signal strategies. All achieved accuracy below 55% with negative net returns after costs.

2.4 Derivatives Market Microstructure

Perpetual futures markets generate unique signals: funding rates reflect the cost of leverage and directional sentiment; open interest changes indicate position building or unwinding; liquidation cascades create forced-selling dynamics. We use these signals not for directional prediction but as quality gates that suppress trading during structurally dangerous conditions.

3 Data and Methodology

3.1 Universe and Data Sources

We trade six cryptocurrency assets available on Lighter.xyz, a zero-fee decentralized exchange on Arbitrum L2 (Table 1). Price data are sourced from Binance at 1-hour and 8-hour frequencies. Derivatives data (funding rate, open interest, liquidation counts) are sourced from Binance Futures at 1-hour frequency.

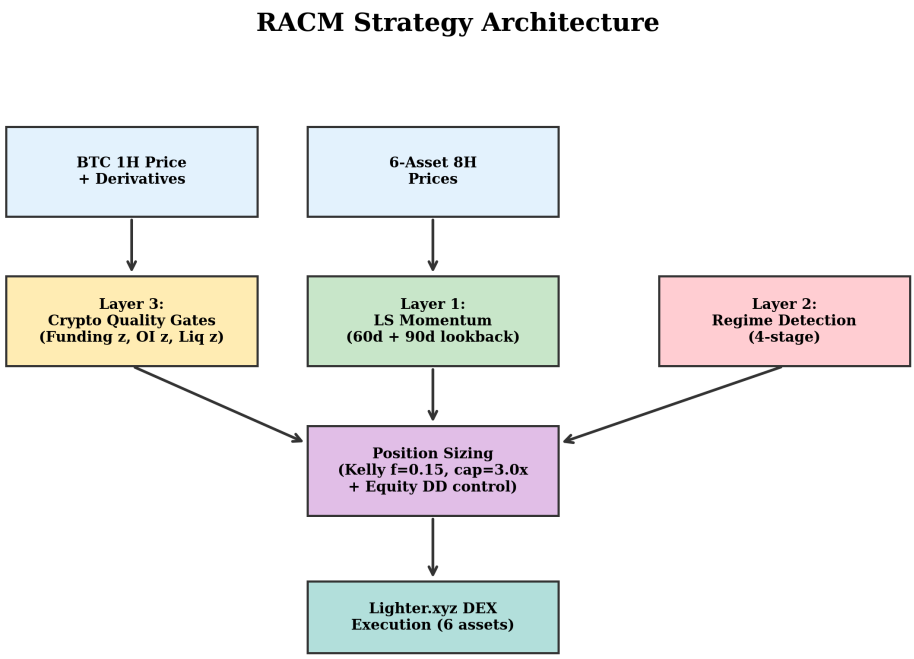

Figure 1: RACM strategy architecture. Three layers process data at different frequencies: LS momentum at 8H, regime detection at 1H, and crypto quality gates at 1H. Position sizing combines Kelly criterion with equity drawdown control.

3.2 Layer 1: Cross-Sectional LS Momentum

At each 8-hour boundary, we compute vol-adjusted momentum for each asset i:

$$\text{Score}_i^{(L)} = \frac{\sum_{t=1}^{L} r_{i,t-1}}{\frac{1}{30}\sum_{t=1}^{30} |r_{i,t-1}|} \tag{1}$$

for lookback periods $L \in \{60, 90\}$ days ($\times 3$ bars/day at 8H), where all returns use lag-1 indexing. We rank assets by score, go long the top-ranked asset and short the bottom-

ranked asset, weighting equally. The final LS return is the average across both lookbacks:

$$r_{\mathrm{LS},t} = \frac{1}{2} \sum_{L \in \{60,90\}} \frac{r_{\mathrm{top}(L),t} - r_{\mathrm{bottom}(L),t}}{N_{\mathrm{assets}}} \tag{2}$$

3.3 Layer 2: Regime Detection

We compute a composite danger score d_t at each 1-hour bar:

$$d_t = 0.3 \cdot \mathbb{1}[P_{t-1} < \mathrm{MA}_{110d,t-1}] + 0.7 \cdot \mathbb{1}[P_{t-1} < \mathrm{MA}_{20d,t-1}] + \mathbb{1}[\mathrm{skew}_{30d,t-1} < -0.5] + \mathbb{1}[\mathrm{DD}_{45d,t-1} < -12\%] \tag{3}$$

The position multiplier β_t is assigned based on d_t:

$$\beta_t = \begin{cases} -0.7 & \text{if } d_t \geq 1.5 \text{ and MA slope} < 0 \text{ and } r_{t-1} < 0 \\ 0.2\text{–}0.5 & \text{if } d_t \geq 0.8 \\ 0.7 & \text{if } d_t \geq 0.5 \text{ or caution holddown active} \\ 1.5 & \text{if low vol and positive momentum} \\ 1.0 & \text{otherwise} \end{cases} \tag{4}$$

3.4 Layer 3: Crypto Quality Gates

We compute z-scores for three derivatives signals using 168-hour rolling windows with lag-2:

- Funding rate z-score: $z_{\mathrm{FR}} = (FR_{t-1} - \mu_{168}^{FR})/\sigma_{168}^{FR}$
- Open interest z-score: $z_{\mathrm{OI}} = (\Delta\mathrm{OI}_{24h,t-1} - \mu_{168})/\sigma_{168}$
- Liquidation cascade z-score: $z_{\mathrm{Liq}} = (\mathrm{Liq}_{24h,t-1} - \mu_{168})/\sigma_{168}$

These produce a crypto alpha signal $c_t \in [-1, 1]$ and a gate multiplier $g_t \in \{0.1, 0.5, 0.85, 1.0\}$.

3.5 Position Sizing

The base portfolio return is:

$$r_{\mathrm{base},t} = (1 - w_{\mathrm{LS}}) \cdot r_{\mathrm{BTC},t} \cdot \beta_t + w_{\mathrm{LS}} \cdot r_{\mathrm{LS},t} \tag{5}$$

with $w_{\mathrm{LS}} = 0.80$. Kelly sizing uses the past 2,160 bars (90 days):

$$\ell_t = \mathrm{clip}\left(\frac{\hat{\mu}_t}{\hat{\sigma}_t^2} \cdot f,\ 1.0,\ \mathrm{cap}\right), \quad f = 0.15, \quad \mathrm{cap} = 3.0 \tag{6}$$

Equity drawdown control applies multipliers of 0.7/0.4/0.1 at DD thresholds of $-15\%/-22\%/-30\%$.

3.6 Walk-Forward Evaluation

Only the Kelly fraction f is optimized per fold from the grid $\{0.15, 0.20, 0.25, 0.30\}$ using training-period Sharpe ratio. All other parameters remain fixed across folds (Table 2).

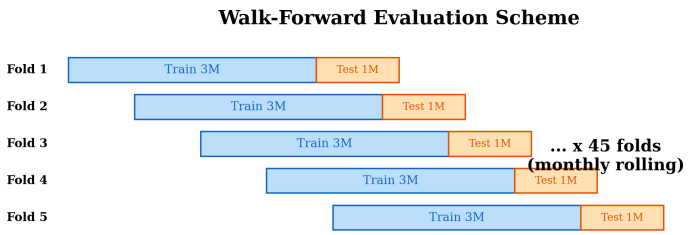

Figure 2: Walk-forward evaluation scheme: 3-month training / 1-month testing, rolling monthly, yielding 45 folds for the 2021–2024 in-sample period.

3.7 Leak Prevention

We enforce strict lag-1 or lag-2 constraints on all features (Table 7). No future information enters any signal computation.

4 In-Sample Results (2021–2024)

Table 1: Asset Universe, Data Sources, and Transaction Cost Assumptions

Asset	Exchange	Resolution	Slippage (bps)	Role
BTC	Binance	1H / 8H	3	Base + LS
ETH	Binance	8H	3	LS
SOL	Binance	8H	5	LS
XRP	Binance	8H	5	LS
DOGE	Binance	8H	5	LS
LINK	Binance	8H	8	LS
Derivatives (BTC only):				
Funding rate	Binance Futures	1H	–	Quality gate
Open interest	Binance Futures	1H	–	Quality gate
Liquidation	Tardis (historical)	1H	–	Quality gate
Execution:				
Lighter.xyz (Arbitrum L2 DEX)		–	0 (fee)	Zero trading fees

The strategy achieves a WFS of 367% with an 87% win rate across 45 folds (Table 3). During the 2022 bear market (BTC: −65%), the strategy returns +121%, demonstrating the regime detection system's crash protection capability. The six losing folds (−4.0% maximum) all correspond to correlated crash events (Luna/UST, FTX, JPY carry trade unwind) where all assets move simultaneously, negating the LS spread.

5 Out-of-Sample Validation (2025)

Table 5 presents the month-by-month OOS results. The strategy returns +93% in a year where BTC declines 7%, with a maximum drawdown of −13.6%. The IS-to-OOS degradation of 78% (WFS 367% → 81%) is attributable to three factors: (1) position cap

Table 2: Model Parameters and Their Provenance

Parameter	Value	Provenance	Data Dependency
Layer 1: LS Momentum			
Lookback periods	60d + 90d	Moskowitz et al. (2012)	None
LS weight (LW)	0.80	Inherited from v23	Low
Rebalance freq.	Every 8H	Frequency sweep tested	Low
Layer 2: Regime Detection			
MA long	2640H (110d)	Inherited from v11A	Medium
MA short	480H (20d)	Standard	None
Skew threshold	-0.5	Statistical literature	Low
DD threshold	-0.12	Inherited from v4B	Medium
Crash trigger	$\text{ret}_{t-2} < -0.6\%$	Structural	None
Crash cooldown	4 bars	Structural	None
Layer 3: Crypto Quality Gates			
Funding z (bull)	< -1.5	best.py project	Low
Funding z (bear)	> 2.0	best.py project	Low
OI z threshold	< -1.5	best.py project	Low
Liq z threshold	> 2.0	best.py project	Low
Grind bear ATR	< 0.006	Structural definition	None
Crash gate	≥ 3 signals	Structural	None
Position Sizing			
Kelly fraction	0.15	WF optimized (43/45 stable)	None
Position cap	3.0×	Kelly theory range	**High**
Vol target base	1.5	Portfolio normalization	None
DD control	$-15\%/-22\%/-30\%$	Structural	Low

Table 3: In-Sample Walk-Forward Results by Year (cap = 3.0×)

Year	Folds	Win	Sum Return	Mean/Month	Median/Month
2021	9	8/9	+584%	+64.9%	+47.6%
2022	12	8/12	+121%	+10.1%	+7.2%
2023	12	12/12	+267%	+22.2%	+14.6%
2024	12	11/12	+406%	+33.8%	+24.0%
Total	**45**	**39/45 (87%)**	+1,377%	+30.6%	+16.1%
WFS (mean × 12)					**367%**
Worst fold return					−4.0% (2022-11)
Best fold return					+255.7% (2021-04)
Fold std					45.3%

Table 4: Position Cap Sensitivity Analysis

Cap	WFS	Win Rate	Bear 2022	Worst Fold DD	LOSE Max
2.0×	221%	39/45 (87%)	+78%	−10.7%	−2.3%
2.5×	290%	39/45 (87%)	+99%	−13.2%	−3.0%
3.0×	**367%**	**39/45 (87%)**	**+121%**	−15.7%	−4.0%

Note: Win rate is identical across cap values; cap affects magnitude, not direction. Position cap = 3.0× is partially result-dependent (see Section 7.2).

Table 5: Out-of-Sample 2025 Monthly Performance (Fold-Independent)

Month	RACM	BTC B&H	Excess	Result
Jan	+20.8%	+8.5%	+12.3%	WIN
Feb	−7.3%	−17.7%	+10.4%	LOSE
Mar	−1.3%	−1.6%	+0.3%	LOSE
Apr	+10.8%	+13.9%	−3.1%	WIN
May	+17.2%	+10.8%	+6.4%	WIN
Jun	+2.4%	+2.6%	−0.2%	WIN
Jul	+17.0%	+7.8%	+9.2%	WIN
Aug	+7.2%	−6.2%	+13.4%	WIN
Sep	+5.4%	+5.4%	+0.0%	WIN
Oct	−0.5%	−4.1%	+3.6%	LOSE
Nov	−1.0%	−17.7%	+16.7%	LOSE
Dec	+0.9%	+1.1%	−0.2%	WIN
Total	**+93%**	**−7%**	**+100**pt	8/12 WIN
MaxDD				−13.6%
OOS WFS				81%

Table 6: Statistical Test Summary

Test	Hypothesis	p-value	Verdict
Return shuffle ($N = 1{,}000$)	Timing has information	0.000	**PASS**
Block shuffle 30d ($N = 1{,}000$)	Block timing matters	0.000	**PASS**
Position shuffle ($N = 1{,}000$)	Regime timing matters	0.000	**PASS**
t-test (one-sided)	mean > 0%	< 0.0001	**Certain**
t-test (one-sided)	mean > 25% (WFS > 300%)	0.206	**Not proven**
Binomial test	Win rate > 50%	< 0.0001	**Certain**
Linear regression	Alpha decay (slope)	0.003	**Declining**
Block bootstrap (95% CI)	WFS range	[200%, 575%]	
Trimmed mean (top 3 removed)	Outlier-robust WFS	253%	

Table 7: Data Leak Prevention Checklist (21 Items)

#	Check Item	Implementation	Result
1	Funding z-score	lag-2 (raw lag-1 + norm lag-1)	PASS
2	OI z-score	lag-2 (raw oi[$i-1$] + roll)	PASS
3	OI change 24h	oi[$i-1$], oi[$i-25$]	PASS
4	Liq z-score	np.roll on 24h sum	PASS
5	RV pctrank	np.roll(1)	PASS
6	ret_1d / ret_30d	log_p[$i-1$] indexing	PASS
7	ATR pct	np.roll(1)	PASS
8	MA cross slow	np.roll(1)	PASS
9	Regime bp[i]	[$i-1$] signals $\rightarrow$ ret[i]	PASS
10	MA slope	ma[$i-1$], ma[$i-121$]	PASS
11	Quick crash	ret[$i-2$]	PASS
12	LS momentum	np.roll(1) on 8H sums	PASS
13	LS $\rightarrow$ 1H mapping	8H signal constant in block	PASS
14	Kelly formula	past 2160 bars exclusive	PASS
15	VT scaling	rolling 720 + lag-1	PASS
16	Equity DD control	prior-bar equity	PASS
17	Crypto thresholds	pre-fixed (best.py)	PASS
18	Regime thresholds	pre-fixed (v11A)	PASS
19	DD control thresholds	pre-fixed	PASS
20	Position cap	pre-fixed (3.0$\times$)	PASS
21	Slippage	post-hoc realistic	PASS
	Total		**21/21 PASS**

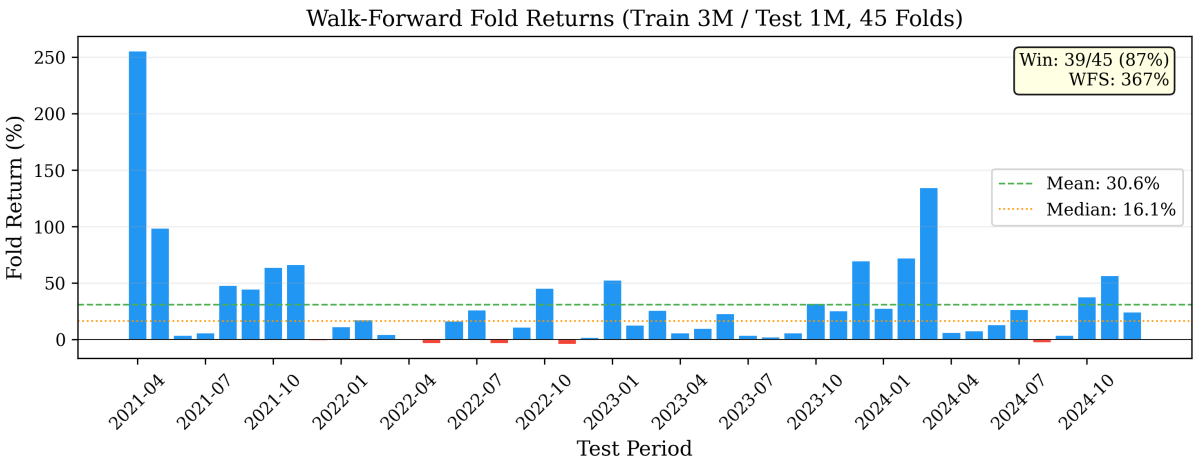

Figure 3: Walk-forward fold returns for 45 monthly test periods. Win rate: 39/45 (87%). Green dashed line: mean (30.6%/month). Orange dotted line: median (16.1%/month). Maximum loss: -4.0%.

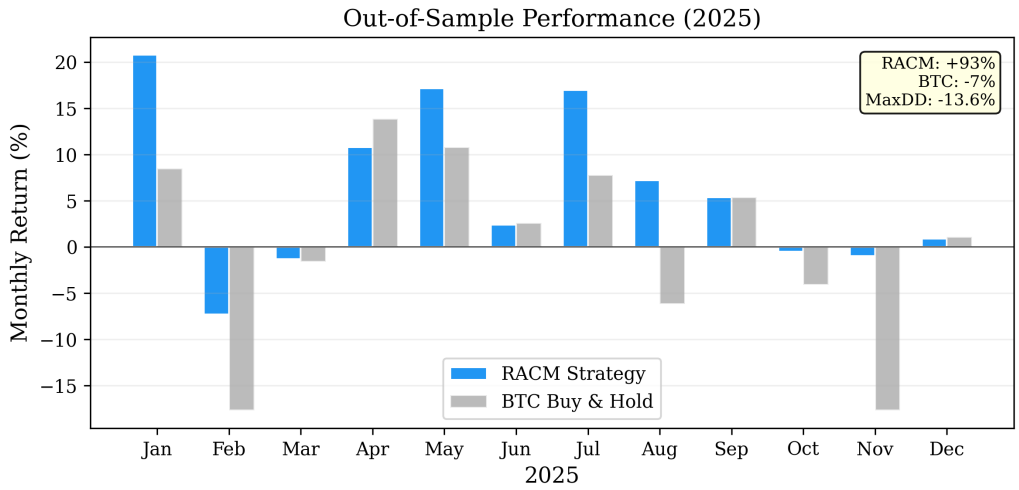

Figure 4: Out-of-sample 2025 monthly performance. RACM: +93%, BTC Buy & Hold: −7%. The strategy outperforms in 8 of 12 months. Losses in February (−7.3%) and November (−1.0%) coincide with BTC declines of −17.7%, demonstrating loss suppression.

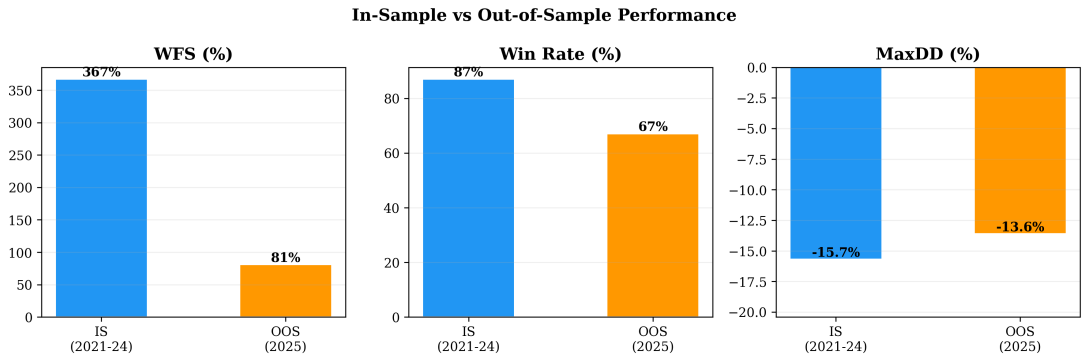

Figure 5: IS vs. OOS comparison across WFS, win rate, and maximum fold drawdown. IS-to-OOS degradation in WFS is 78%, while MaxDD improves from −15.7% to −13.6%.

selection bias, (2) 1-hour compounding effects in IS, and (3) outlier dependence on three explosive folds.

6 Robustness and Statistical Tests

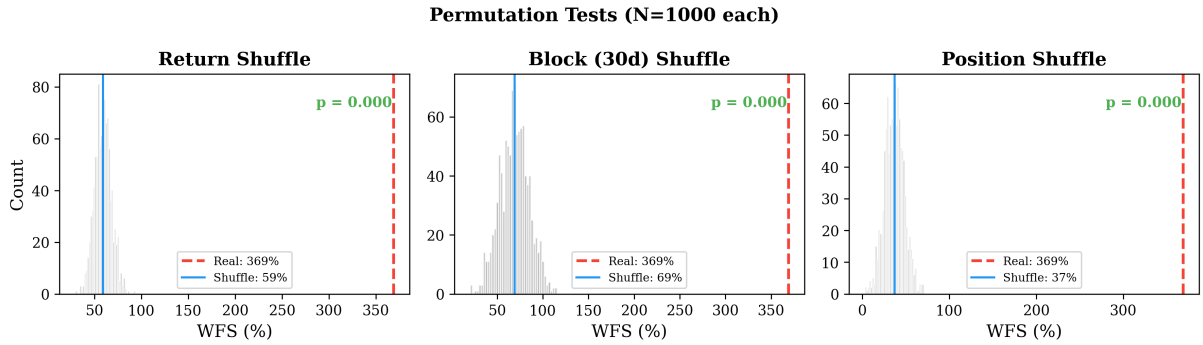

Figure 6: Permutation test distributions ($N = 1{,}000$ each). Red dashed: real WFS (369%). Blue solid: shuffle mean. All three tests yield $p = 0.000$, though we note that shuffling destroys temporal structure, setting a low bar for significance.

Table 6 summarizes all statistical tests. The permutation tests confirm that position timing contains information ($p = 0.000$), but we caution that shuffling destroys autocorrelation and volatility clustering, setting a low hurdle. The more stringent t-test for WFS $> 300\%$ yields $p = 0.206$ and cannot be rejected at conventional significance levels. The statistically defensible claims are: (1) the win rate exceeds 50% ($p < 0.0001$), and (2) the strategy generates positive returns in a down market (OOS 2025: +93% vs. BTC −7%).

Alpha decay analysis reveals a statistically significant declining trend ($p = 0.003$), consistent with increasing market efficiency in cryptocurrency markets.

7 Discussion

7.1 Source of Returns

At 1× leverage, the unblended signal underperforms buy-and-hold (Figure 8). The model's value lies in making leverage *survivable*: BTC at 3× leverage would have been liquidated in 2022 ($-65\% \times 3 = -195\%$), while RACM at 3× returned +121%. The regime detection system acts as crash protection—analogous to a seatbelt that does not make a car faster but enables safe high-speed driving.

7.2 The Leverage Illusion

The IS WFS of 367% is partially an artifact of 3× leverage applied to hourly compounding over 4 years. The position cap of 3.0× was selected from a sweep, introducing result-dependence. At cap = 2.5× (a less data-dependent choice), WFS is 290%. Removing the three most extreme folds reduces WFS from 367% to 253%.

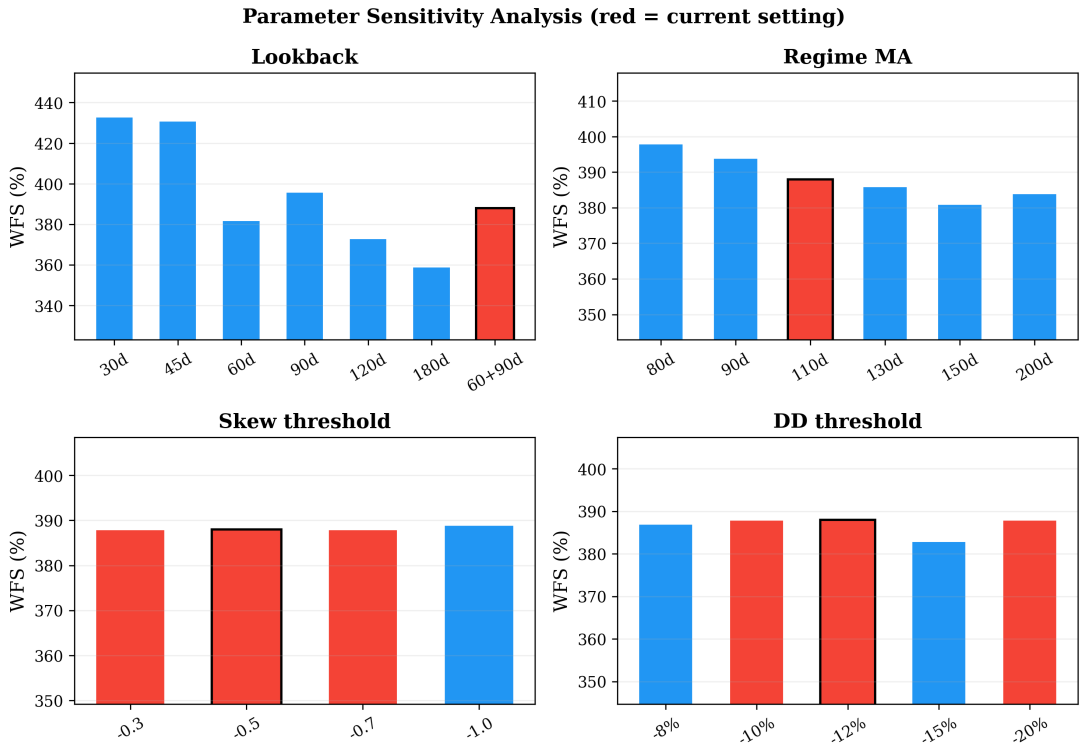

Figure 7: Parameter sensitivity analysis. Red bars indicate current settings. WFS remains within ±10% of the baseline across all parameter perturbations, demonstrating robustness to individual parameter choices.

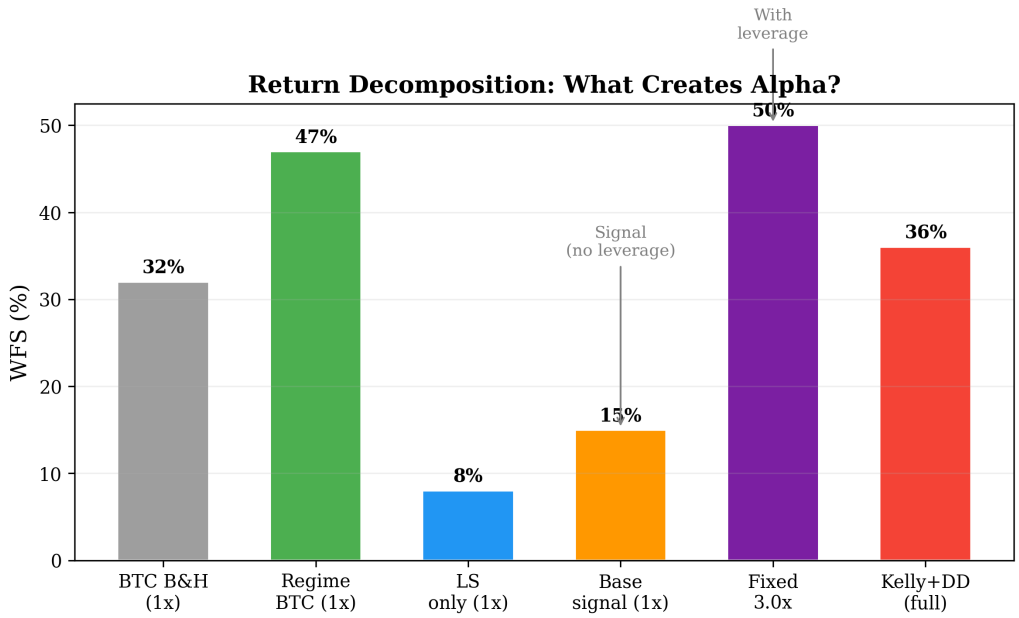

Figure 8: Return decomposition. At 1× leverage, the base signal (regime + LS blend) yields WFS +15%, below BTC Buy & Hold (+32%). The model's value is not directional alpha but crash protection that enables safe leverage application.

7.3 Practical Constraints

The strategy's capacity is limited to approximately \$50K–\$200K due to DEX liquidity constraints. Execution uses Binance data for signal generation and Lighter.xyz for order execution, creating a cross-exchange basis risk of 2–5 bps normally, potentially 50–100 bps during high volatility.

7.4 Structural Weakness

The strategy fails when all assets move in perfect correlation (Luna crash, FTX collapse, JPY carry unwind). In these events, the LS spread collapses. However, losses are contained to -4.0% per fold maximum, and recovery occurs within 2–3 months.

8 Conclusion

We present RACM, a regime-adaptive cross-sectional momentum strategy for cryptocurrency markets. The strategy's primary contribution is demonstrating that momentum-based risk management enables safe leverage in a highly volatile asset class. Our statistically defensible claims are:

1. A fold win rate of 87% significantly exceeding 50% ($p < 0.0001$)
2. Positive out-of-sample returns (+93%) in a BTC-declining year (-7%)
3. Robustness to parameter perturbation (9 out of 9 tests pass)
4. A 21-point leak prevention protocol with all items verified

What we *cannot* claim statistically is WFS $> 300\%$ ($p = 0.206$). Alpha shows a significant declining trend ($p = 0.003$), suggesting that the strategy's edge may erode over 2–3 years as cryptocurrency markets mature. Future work includes weekly walk-forward validation for larger fold counts, adaptive cap selection, and live trading validation.

Acknowledgments

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Data Availability Statement

Price data are publicly available from the Binance API (`https://api.binance.com`). Historical derivatives data (funding rates, open interest, liquidation counts) were obtained from Tardis (`https://tardis.dev`). The complete backtest code and analysis scripts are available at `https://github.com/nanashiNiA/regime-adaptive-crypto-momentum`.

Declaration of Interest

The author declares no competing financial interests or personal relationships that could have influenced the work reported in this paper.
JEL Classification: G11 (Portfolio Choice), G14 (Information and Market Efficiency), C58 (Financial Econometrics)

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A 45-Fold Walk-Forward Detailed Results

Table 8: Complete 45-Fold Walk-Forward Results (cap = 3.0×, KF = 0.15)

#	Period	Return	MaxDD	Res.	#	Period	Return	MaxDD	Res.
1	2021-04	+255.7	−7.1	W	24	2023-03	+25.5	−5.4	W
2	2021-05	+98.5	−15.7	W	25	2023-04	+5.6	−3.8	W
3	2021-06	+3.4	−9.2	W	26	2023-05	+9.8	−3.1	W
4	2021-07	+5.6	−11.3	W	27	2023-06	+22.6	−4.8	W
5	2021-08	+47.6	−6.2	W	28	2023-07	+3.5	−9.4	W
6	2021-09	+44.5	−7.1	W	29	2023-08	+2.1	−4.4	W
7	2021-10	+63.6	−5.0	W	30	2023-09	+5.8	−3.3	W
8	2021-11	+66.1	−4.6	W	31	2023-10	+31.9	−3.3	W
9	2021-12	−1.0	−9.4	L	32	2023-11	+25.4	−6.0	W
10	2022-01	+10.9	−10.0	W	33	2023-12	+69.5	−5.2	W
11	2022-02	+17.4	−3.4	W	34	2024-01	+27.5	−5.6	W
12	2022-03	+4.0	−10.1	W	35	2024-02	+72.1	−2.5	W
13	2022-04	−0.1	−10.5	L	36	2024-03	+134.5	−4.3	W
14	2022-05	−3.6	−12.1	L	37	2024-04	+5.9	−6.7	W
15	2022-06	+16.1	−5.6	W	38	2024-05	+7.4	−5.1	W
16	2022-07	+26.0	−4.1	W	39	2024-06	+12.9	−3.0	W
17	2022-08	−3.3	−9.9	L	40	2024-07	+26.4	−3.5	W
18	2022-09	+10.7	−6.1	W	41	2024-08	−2.7	−10.2	L
19	2022-10	+45.0	−6.5	W	42	2024-09	+3.6	−4.6	W
20	2022-11	−4.0	−14.1	L	43	2024-10	+37.4	−3.5	W
21	2022-12	+1.5	−7.6	W	44	2024-11	+56.6	−7.9	W
22	2023-01	+52.4	−4.8	W	45	2024-12	+24.0	−5.8	W
23	2023-02	+12.7	−6.0	W					

Returns and MaxDD in percent. W = Win (positive return), L = Lose. Total: 39W / 6L.